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Group No: C2

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Title: DeployForge: An Intelligent Hybrid Deployment Orchestration Platform for Containerized and Non-Containerized Applications

Project Proposal

Abstract:

Deploying modern web applications requires developers to manually provision servers, install runtimes, configure reverse proxies, manage SSL certificates, and orchestrate multi-service dependencies a process that is time-consuming and error-prone, especially for students and early-stage developers. Existing platforms such as Heroku and Vercel simplify this but are largely proprietary, cost-prohibitive at scale, and offer limited transparency into how deployment decisions are made. This project proposes DeployForge, a self-hosted deployment automation platform that intelligently deploys both containerized and non-containerized applications from a connected GitHub repository. The system automatically detects whether a repository includes Docker configuration; if present, it deploys via Docker/Docker Compose, respecting the user's setup. If absent, it identifies the application framework (Next.js, React, Express, FastAPI, Django, etc.) and applies a pre-configured buildpack to build and run the application without requiring the user to write any container configuration. The platform further automates infrastructure provisioning (VPS creation via cloud provider APIs), reverse proxy setup, subdomain assignment, SSL certificate issuance, and continuous redeployment on Git push reducing what is traditionally a multi-hour, multi-tool DevOps workflow to a single click. By combining GitHub integration, repository analysis, containerized and non-containerized build pipelines, and automated infrastructure orchestration into one cohesive system, DeployForge demonstrates a practical, extensible approach to modern deployment automation suitable for individual developers, small teams, and educational environments.



Objectives:

1. Design and implement a GitHub OAuth integration allowing users to authenticate and select repositories for deployment without manual code upload.
2. Build a repository analysis engine that reliably detects the presence of Docker configuration and, in its absence, correctly identifies the application framework/stack.
3. Develop a dual build pipeline a Docker execution engine for containerized projects and a buildpack-based engine for non-containerized projects (Next.js, React, Express, FastAPI, Django, etc.).
4. Implement an automated infrastructure provisioning module that creates and configures compute resources via cloud provider APIs, including Docker installation, reverse proxy setup, and SSL automation.
5. Build a dynamic subdomain and routing system that assigns each deployed application a unique subdomain under a base domain with automatic HTTPS.
6. Implement continuous deployment via webhooks, so that every push to the connected repository triggers an automatic rebuild and redeploy, along with basic logging and monitoring of deployment status.

Technology Stack:

Frontend: React / Next.js dashboard for repository selection, deployment status, and logs.

Backend: Node.js (Express) / Python (FastAPI) API server orchestrating the deployment pipeline.

Authentication: GitHub OAuth Apps and Webhooks API.

Containerization: Docker, Docker Compose.

Orchestration: Docker Engine API (Phase 1); Kubernetes (Phase 2, for scaling).

Infrastructure Provisioning: DigitalOcean API / AWS SDK.

Reverse Proxy: Nginx or Traefik (for dynamic subdomain routing).

SSL: Let's Encrypt via Certbot or Traefik's built-in ACME support.

Database: PostgreSQL user data, deployment records, and logs.



Queue / Background Jobs: Redis with BullMQ (or Celery for a Python backend).

Monitoring & Logs: Real-time log streaming to the dashboard; Grafana/Loki as a future enhancement.

Scope:

In Scope: GitHub OAuth login and repository listing; Dockerfile/Compose detection and execution; framework auto-detection for common stacks (Next.js, React, Express, FastAPI, Django); automated VPS provisioning on one cloud provider; reverse proxy, subdomain, and SSL automation; automatic redeployment on Git push via webhooks; and a basic deployment dashboard with logs and status.

Future Scope: migration to a Kubernetes-based architecture for true auto-scaling and multi-tenant isolation; one-click managed service attachments (Postgres, Redis, RabbitMQ) with auto-wired environment variables; multi-cloud provisioning across AWS, GCP, and Azure; an agentic AI layer that diagnoses build/deployment failures from logs and proposes or applies fixes autonomously; a community-driven marketplace of buildpacks and templates; and role-based access control for multi-developer teams.

SDGs Mapped with Explanation:

SDG 9 - Industry, Innovation and Infrastructure: The project contributes to resilient digital infrastructure by simplifying and automating the deployment pipeline, promoting innovation in software delivery practices and enabling more developers to build and ship applications reliably.

SDG 4 - Quality Education: By automating complex DevOps tasks (server provisioning, containerization, SSL, deployment), the platform lowers the technical barrier for students and beginner developers, allowing them to learn and deploy real-world applications without needing deep infrastructure expertise.

Inclusion of Benchmark parameters as bellow:



(choose whatever is feasible & possible for your group as per the topic selected)

- A. Requirement Gathering ☒
- B. Orchestration of System Scalability & Load Balancing ☒
- C. Database Scaling ☐
- D. Orchestration of Caching Strategies for Performance Improvement through Redis / Memcache ☐
- E. Orchestration of Message Queues & Buffering through RabbitMQ / Kafka ☐
- F. Orchestration of Microservices & API Gateway ☐
- G. Orchestration of Cloud Usage for Deployment / Service Usage ☒
- H. Applicability of Content Delivery Network (CDN) ☐
- I. Orchestration of Authentication Services ☒
- J. Orchestration of AI Use Case ☐
- K. Orchestration of Data Analytics Use Case ☒
- L. Orchestration of Monitoring & Logging, Notification System ☒
- M. Containerisation of Entire Project & Developing CI/CD Pipeline ☒